THE UNIVERSITY OF HONG KONG

BACHELOR OF ENGINEERING EXAMINATION

DEPARTMENT OF CIVIL ENGINEERING

CIVL2108: PRINCIPLES OF CIVIL ENGINEERING MANAGEMENT

May 17, 2024 9:30AM - 12:30PM

Answer ALL questions.

A compound interest table is provided.

Use of Calculators:

Only approved calculators as announced by the Examinations Secretary can be used in this examination. It is candidates' responsibility to ensure that their calculator operates satisfactorily, and candidates must record the name and type of the calculator used on the front page of the examination script.

1. Northern Metropolis is positioned as a new economic engine for Hong Kong, a base for new industries, and a platform for high-quality cooperation with the rest of the Greater Bay Area. The new development land will provide over 500 000 new residential units and accommodate 2.5 million people. Northern Metropolis could be an ideal home for circularity. Assuming you are a project manager for the housing projects in Northern Metropolis, please explain to the stakeholders.
   a) the concept of circular economy (5 marks)
   b) the benefit of adopting circularity in civil engineering projects (5 marks)
   c) the changes in project management if adopting circularity. (10 marks)

2. The following table shows the duration and the logical relationship of the activities involved in a construction project:

Activity ID	Predecessor	Duration (days)
A	-	8
B	A	5
C	B	12
D	B	7
E	B	9
F	B	18
G	D	13
H	E	10
I	F,G,H	9
J	I	8
K	C,J	12

   a) Prepare an activity-on-node network diagram based on the information given in the table above. (5 marks)
   b) Prepare a table showing ES, EF, LS, and LF of each activity. (10 marks)
   c) Please identify the critical path and the free floats for each activity in the chart. (5 marks)
   d) With the development of new technologies, the duration of activity G, H and F will be reduced to 5 days, 3 days and 11 days respectively if an additional activity L is carried out after activity B and before the commencement of activity G, H and F. The duration of activity L is 3 days. Assuming you are the project manager, will you adopt the new technology? How many days can be saved in total for the whole project? Please explain and illustrate with diagram. (10 marks)

3. On the first day of April, the Zhuhai port of the Hong Kong-Zhuhai-Macao Bridge (HZMB) saw over 19,000 inbound and outbound vehicles, making a record high since its opening. The 55-km Bridge, comprising the 12km Hong Kong Link Road, 29.6km Main Bridge and

13.4km Zhuhai Link Road, is the longest bridge-cum-tunnel sea crossing in the world. Operating 24 hours a day, HZMB puts major cities in the Pearl River Delta within a three hours' commute from Hong Kong; and it will take only 40 minutes to travel the distance of approximately 42km from Hong Kong Port to Zhuhai Port and Macao Port.

a) Please identify the social benefits and social costs of HZMB. (10 marks)
b) Please list the roles and responsibilities of all involvements during the whole life cycle of the project. (15 marks)

4. The construction cost of a sewerage system is estimated to be $30,000,000. The annual operation, maintenance and repair (i.e., OMR) cost will be $1,000,000 per year. The annual income (i.e. benefit) from the collection of sanitation fees from the users will be $3,500,000. Taking a time horizon of 30 years and a discount rate of 10%,
   a) Determine if the project is worthwhile. (10 marks)
   b) After a detailed investigation, it is found out that annual income will increase 100% after 10 years' operation due to the completion of new buildings nearby. Please determine if the project is worthwhile. (15 marks)

Compound Interest Table

$i = 10\%$

n	caf $(1+i)^n$	pwf $\frac{1}{(1+i)^n}$	uscaf $\frac{(1+i)^n-1}{i}$	ussff $\frac{i}{(1+i)^n-1}$	uspwf $\frac{(1+i)^n-1}{i(1+i)^n}$	uscrf $\frac{i(1+i)^n}{(1+i)^n-1}$
1	1.1000	0.9091	1.0000	1.0000	0.9091	1.1000
2	1.2100	0.8264	2.1000	0.4762	1.7355	0.5762
3	1.3310	0.7513	3.3100	0.3021	2.4869	0.4021
4	1.4641	0.6830	4.6410	0.2155	3.1699	0.3155
5	1.6105	0.6209	6.1051	0.1638	3.7908	0.2638
6	1.7716	0.5645	7.7156	0.1296	4.3553	0.2296
7	1.9487	0.5132	9.4872	0.1054	4.8684	0.2054
8	2.1436	0.4665	11.436	0.0874	5.3349	0.1874
9	2.3579	0.4241	13.579	0.0736	5.7590	0.1736
10	2.5937	0.3855	15.937	0.0627	6.1446	0.1627
11	2.8531	0.3505	18.531	0.0540	6.4951	0.1540
12	3.1384	0.3186	21.384	0.0468	6.8137	0.1468
13	3.4523	0.2897	24.523	0.0408	7.1034	0.1408
14	3.7975	0.2633	27.975	0.0357	7.3667	0.1357
15	4.1772	0.2394	31.772	0.0315	7.6061	0.1315
16	4.5950	0.2176	35.950	0.0278	7.8237	0.1278
17	5.0545	0.1978	40.545	0.0247	8.0216	0.1247
18	5.5599	0.1799	45.599	0.0219	8.2014	0.1219
19	6.1159	0.1635	51.159	0.0195	8.3649	0.1195
20	6.7275	0.1486	57.275	0.0175	8.5136	0.1175
21	7.4002	0.1351	64.002	0.0156	8.6487	0.1156
22	8.1403	0.1228	71.403	0.0140	8.7715	0.1140
23	8.9543	0.1117	79.543	0.0126	8.8832	0.1126
24	9.8497	0.1015	88.497	0.0113	8.9847	0.1113
25	10.835	0.0923	98.347	0.0102	9.0770	0.1102
26	11.918	0.0839	109.18	0.0092	9.1609	0.1092
27	13.110	0.0763	121.10	0.0083	9.2372	0.1083
28	14.421	0.0693	134.21	0.0075	9.3066	0.1075
29	15.863	0.0630	148.63	0.0067	9.3696	0.1067
30	17.449	0.0573	164.49	0.0061	9.4269	0.1061

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